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2016-01-1495
Published 04/05/2016
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doi:10.4271/2016-01-1495
saetransaf.saejournals.org

Event Data Recorder (EDR) Developed by Toyota Motor Corporation

Motomi Iyoda
Toyota Motor Corporation

Tom Trisdale
Toyota Motor Sales USA Inc.

Rini Sherony and Daniel Mikat
TEMA

William Rose
Bosch Automotive Service Solutions Inc.

ABSTRACT

An event data recorder (EDR) records the vehicle status at the timing of an accident. Toyota Motor Corporation began the sequential introduction of EDRs onto its vehicles from August 2000. Currently, about 70% of all Toyota's vehicles in North America are equipped with an EDR, which is more than the average rate of EDR installation in vehicles in North America (around 50%). The U.S. has introduced regulations for EDRs. Toyota regards these as minimum requirements and also records additional data for accident analysis, including the following: (1) pre-crash data, (2) side crash data, (3) rollover data, (4) pedestrian protection pop-up hood (PUH) data, and (5) vehicle control history (VCH) data from a non-crash triggered recording system. The regulations stipulate that EDR data retrieval must be possible using a commercially available tool. The developed system uses the Crash Data Retrieval (CDR) tool manufactured by Bosch. Data from all Toyota EDRs around the world can be retrieved using the CDR tool, including in North America. Since EDR data collection and analysis are very important for real-world accident analysis, an automatic online EDR data collection system was constructed with the cooperation of Bosch. Toyota is currently developing new systems using EDR data. As an example, making effective use of crash severity ΔV and seatbelt data, the Advanced Automatic Collision Notification (AACN) system is due to start trials in Japan from November 2015. The expansion of EDR functions and system development using EDR data are likely to continue growing in importance.

CITATION: Iyoda, M., Trisdale, T., Sherony, R., Mikat, D. et al., "Event Data Recorder (EDR) Developed by Toyota Motor Corporation," *SAE Int. J. Trans. Safety* 4(1):2016, doi:10.4271/2016-01-1495.

1. INTRODUCTION

Passive safety technology has made substantial advances in recent years and technology is being developed to protect occupants in a variety of crash types. Systems targeting frontal and side crashes, as well as rollovers are steadily becoming standard on vehicles. As well as various types of airbags and other occupant restraint systems, vehicles are also provided with a range of airbag control sensors that determine the optimum airbag deployment timings. It is possible to identify the type and severity of accidents by recording data from these sensors when a crash occurs. Data can also be recorded from the local area network (LAN) of the vehicle to identify the vehicle status leading up to the crash to determine the details of the vehicle before the accident.

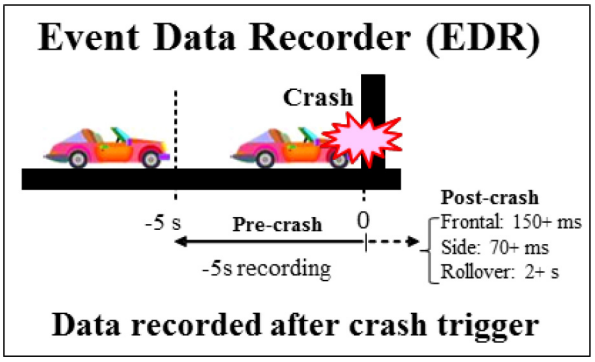


Figure 1. Event data recorder.

An event data recorder (EDR) records the vehicle status when an accident occurs.

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Recording starts with the acceleration (G) at the beginning of the crash judged as almost time zero. The recorded data consists of precrash data, which details the vehicle status before the crash, and post-crash data, which details the occupant status from the start of the crash and the severity of the crash.

The pre-crash data shows the vehicle status and driving operations. Data, such as the vehicle speed, engine speed, accelerator pedal position, and brake status is recorded at 0.5 or 1-second intervals. Since these values vary in accordance with the driving operations, new data must be constantly recorded.

Post-crash data includes airbag sensor data, which indicates the type and severity of the crash, the occupant status (such as whether seatbelts are being worn), and airbag deployment data.

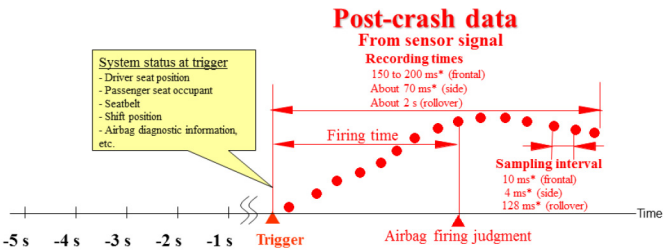
Since the recorded data describes driving and accident conditions, the property rights of EDR data belong to the owner. Therefore it is necessary to respect privacy and obtain the permission of the user before downloading data. It is important to record the most effective data for accident analysis more closely while ensuring the accuracy and reliability of the data.

2. EDRs DEVELOPED BY TOYOTA

2-1 Prior to EDR Regulations

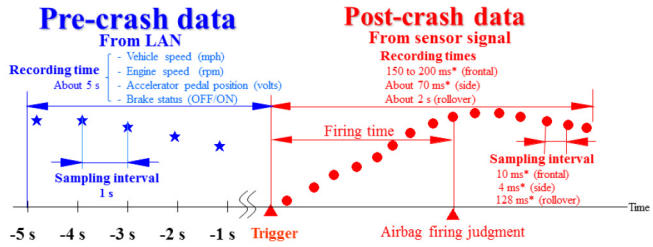
Toyota began the sequential adoption of EDRs onto its vehicles in August 2000. To obtain pre-crash data of the vehicle status from other systems, the vehicle must have a LAN. In Toyota the presence of a LAN determines whether pre-crash data can be obtained .

Toyota developed Type A and B EDRs. Type A records only post-crash data in vehicles without a LAN and Type B records both preand post-crash data.



Note 1: Frontal crash data is recorded by the Type A EDR and is available in all models. Side crash and rollover data depends on the model.

Figure 2. Type A: post-crash data only.



Note 1: The pre-crash data specifications recorded by the Type B EDR are common to all models.
Note 2: Frontal crash data is also recorded by the Type B EDR, and is available in all models. Side crash and rollover data depends on the model.

Figure 3. Type B: pre- and post-crash data.

APPENDIX A shows the year of EDR introduction for each model in North America, highlighting the timing of Type B (pre-crash data) adoption. Although the details of the recorded data differ between models, all models in North America were equipped with pre-crash data by the end of 2010.

In around 2009, more than 150 EDR data download tools (also called the read-out tools (ROTs), see section 2-3) were delivered in North America, and a thorough field investigation using EDR data was started in May of the same year. As shown in Fig. 4, the data identified cases caused by the driver mistakenly depressing the accelerator instead of the brake.

Toyota delivered a total of 230 EDR download tools around the world and began responding to inquiries from the field. At the same time, download tools were provided to the National Highway Traffic Safety Administration (NHTSA) in the U.S. for cooperation with an investigation by that agency.

The interim report issued by NHTSA on August 10, 2010, stated that EDR data was useful in around 90% of cases of possible sudden acceleration and that the main causes were mistaken pedal application and late braking (Fig. 5).

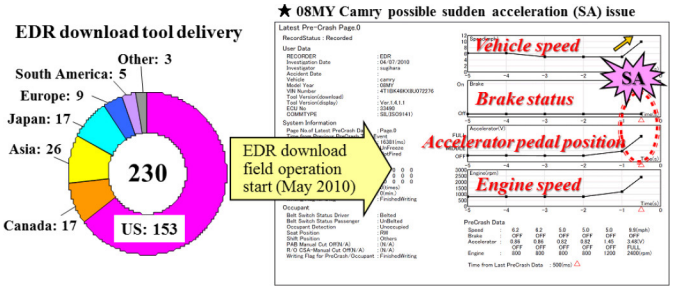
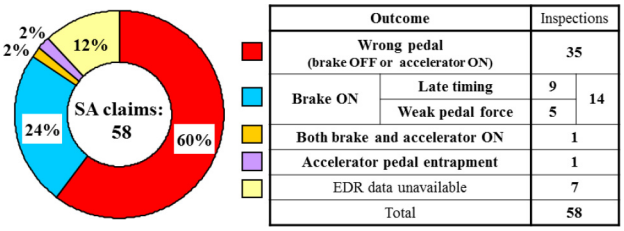


Figure 4. EDR download tool delivery.



Outcome		Inspections	
Wrong pedal (brake OFF or accelerator ON)		35	
Brake ON	Late timing	9	14
	Weak pedal force	5	
Both brake and accelerator ON		1	
Accelerator pedal entrapment		1	
EDR data unavailable		7	
Total		58	

Figure 5. NHTSA investigation of possible sudden acceleration.

2-2. Compliance with EDR Regulations

Using the experience obtained from the field investigations of possible sudden acceleration, the EDR recording specifications were substantially revised to enable the recording of useful data for accident analysis. This revision coincided with the enactment of EDR regulations in the U.S. [1] Regarding these regulations as minimum requirements, it was decided to add various types of data to enable accident analysis (although this data was partially recorded prior to the enactment of the EDR regulations, the following sections describe the differences with the regulations). The regulatory-compliant EDR, called the 12EDR, was phased in starting in 2011 and was in all vehicles prior to the regulation effective date of Sept. 2012. (the two digits before “EDR” indicate the year of market launch).

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2-2-1. Addition of Pre-Crash Data

The principle of data capture was revised to enable the recording and accurate identification of driver operations and the vehicle status before the crash (Fig. 6).

Previously, only the minimum amount of data was recorded, including vehicle speed, engine speed, accelerator pedal position, and brake status. The necessary data was selected after considering the driver operations and inputs relevant to the three overall vehicle states (running, stopping, and turning) and listing the vehicle state outputs changed by these inputs. The data that was added is shown by the large black circles (●) in Figure 6. Figure 7 outlines the details of the regulatory-compliant 12EDR. Major enhancements include the increased pre-crash data, and the shorter sampling time (0.5 s, reduced from 1.0 s). Excluding the frontal crash recording time, the post-crash data specifications are virtually unchanged.

➢ **Focus on Running, Stopping, and Turning**

➢ **Record both Driver Inputs and Vehicle Status**

	Driver operation/input	Vehicle status/output
Running	*- Accelerator pedal position* - Shift position ● **Cruise control (ON/OFF), drive mode switch**	*- Vehicle speed* *- Engine speed* ● **Throttle angle** ● **Low longitudinal acceleration (Gx)**
Stopping	*- Brake status (ON/OFF)* ● **Master cylinder pressure**	
Turning	● **Steering angle**	● **Yaw rate**

__: Data required by Part 563 regulations

● :Toyota enhanced data (from 06EDR)

Figure 6. Pre-crash data capture principle.

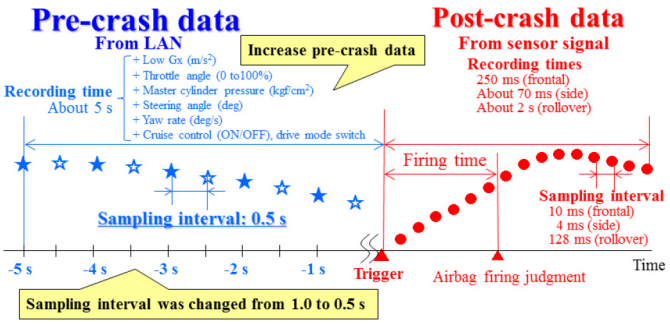


Figure 7. 12EDR implementation.

The following test demonstrates the enhancement in recording accuracy achieved by 12EDR. Figure 8 shows an example of the data recorded by 12EDR for a short accelerator ON operation (less than 1 s) after changing the sampling time.

♦ ***Test: short accelerator ON***

◆**Test: short accelerator ON**

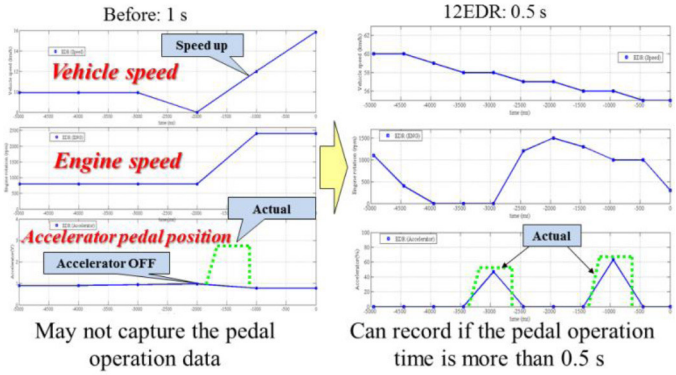


Figure 8. Sampling time change.

Figure 9 shows a recording of the operation of the brake override system (BOS), which reduces the throttle angle and decreases the vehicle speed when the accelerator and brake pedals are depressed simultaneously.

♦ ***Test: both pedals applied***

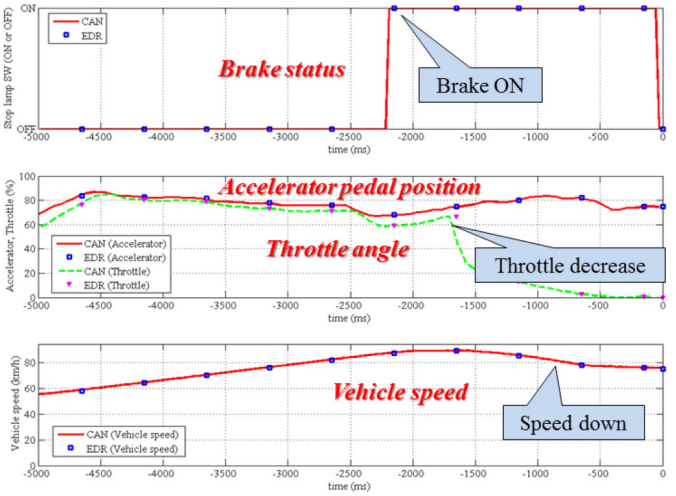


Figure 9. BOS recording.

2-2-2. Post-Crash Data Required for Crash Reconstruction

Frontal Crash

The change in longitudinal velocity (also called “delta-V, longitudinal,” or ΔVx) is recorded in accordance with the requirements for continuously running algorithms described in the regulations.

To identify ΔVx more precisely, the recorded value is calculated using the moving average of the value sampled from the longitudinal acceleration (Gx) sensor 20 times over a period of 10 ms at an interval of 0.5 ms (Fig. 10).

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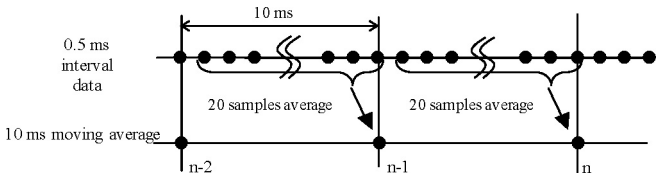


Figure 10. Frontal crash recorded data.

The calculated ΔVx is recorded every 10 ms over a period of 250 ms.

Recording is ended when the "end of event" conditions are satisfied. The purpose of this process is to end recording when the crash is over and to prepare for the next recording. It enables the accurate identification of severe crashes such as multiple impacts (Fig. 11).

Since the data length depends on the end of event timing, the number of the recorded data is also recorded. If ΔVx or the data number cannot be recorded because the crash has disconnected the battery, the state of the ignition power source (i.e., the battery state) is attached to the front part of each data item to clarify the effective range of the recorded data. This is for more severe crash than FMVSS208. The CDR tool reports ON/OFF state in the delta V time series table.

To prevent unnecessary recording triggered by rough roads, slight crashes, and the like, the regulations require an integral guard value (trigger threshold: moving average over 150 ms (V150) ≥8 km/h) to be set.

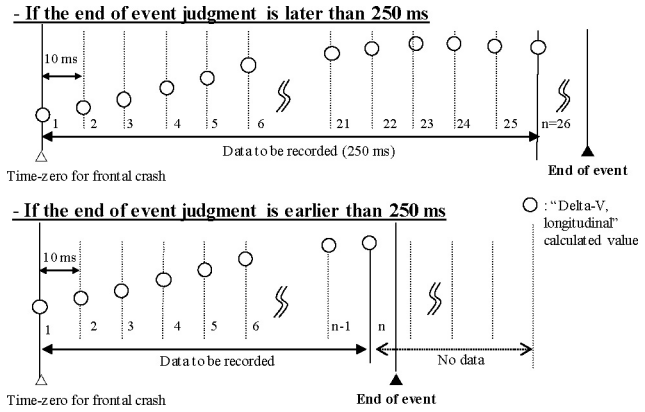


Figure 11. Frontal crash buffer data.

2-2-2-2 Additional Side Crash Data Recording

In a side crash, the values from the lateral acceleration (Gy) sensors installed at the B-pillar, C-pillar, and airbag electronic control unit (ECU) are recorded (Fig. 12). Figure 13 analyzes a rear-seat pole side crash at 29 km/h. Because large rotational movement is generated in the vehicle, the airbag ECU Gy sensor can only detect an integral change in lateral velocity ("delta-v, lateral," or ΔVy) of half that value. The regulations only require the recording of ΔVy inside the airbag ECU. However, it was concluded that the EDR should also record the B- and C- pillar data in the event of a crash outside the center of gravity of the vehicle [2,3].

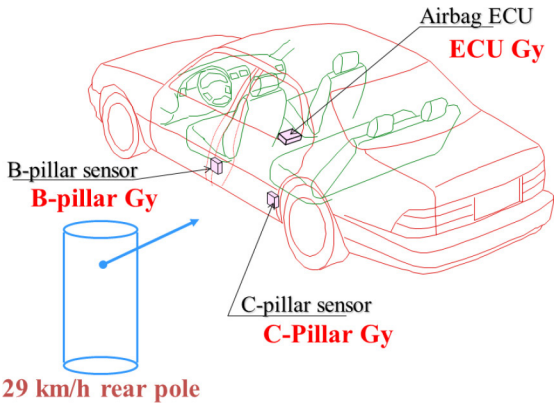


Figure 12. Side airbag sensing system.

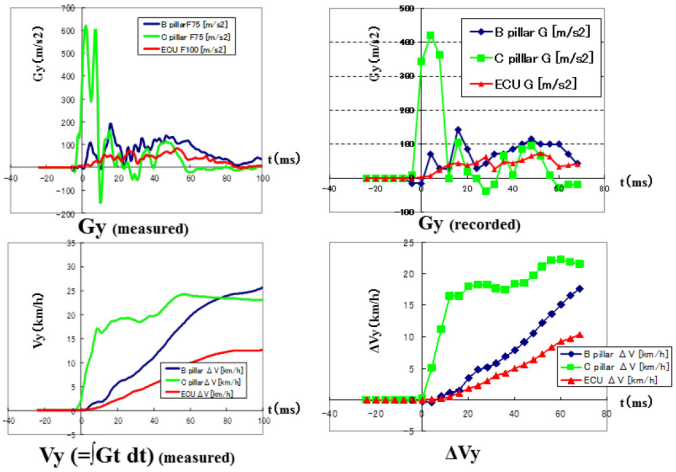


Figure 13. 29 km/h rear pole.

To identify the ECU Gy, B-pillar Gy, and C-pillar Gy more precisely, the recorded values are calculated using the moving average of each Gy sensor value sampled 8 times over a period of 4 ms at an interval of 0.5 ms (Fig. 14).

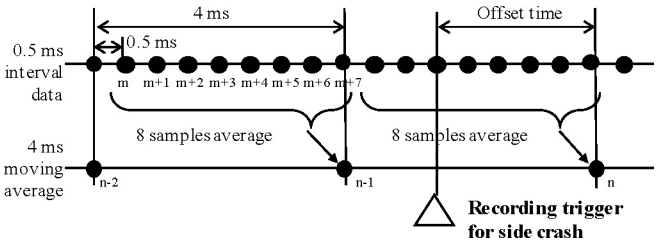


Figure 14. Additional side crash recorded data.

Gy is recorded over a period of 96 ms at an interval of 4 ms.

The side crash judgment trigger is performed using the signals from each sensor. However, since the longitudinal position of each sensor differs, the start-up timing of each crash Gy also changes. Therefore, the fastest data is used as the side crash trigger data. Since the location and trigger threshold of each sensor differ, it is necessary to record the Gy before the trigger. The Gy of each side crash sensor is then recorded over a period from -24 ms to 72 ms based on the fastest trigger.

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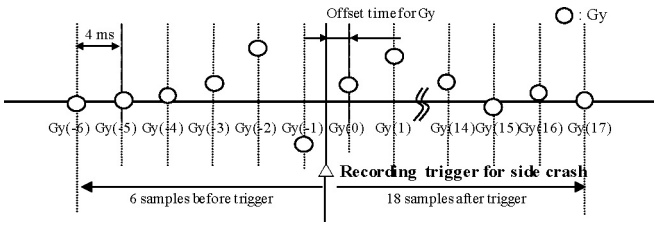


Figure 15. Additional side buffer data (for ECU Gy, B-pillar Gy, and C-pillar Gy).

It should also be noted that the same recording as for the B-pillar sensor is carried out if a front door G sensor is provided in addition to the side crash B-pillar sensor.

Rollover Record

There are two main categories of rollover events, with and without a crash (Fig. 16) [2,3].

The non- crash type tends to be a slow screwing rollover in which the rotation of the vehicle follows an inclined surface. The crash type tends to involve the vehicle tripping over a curb, causing the vehicle to move sideways and rotate quickly.

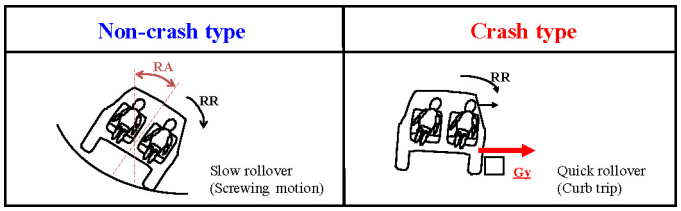


Figure 16. Rollover event types.

The rollover judgment trigger is a roll rate (RR) over 43 deg/s or a roll angle (RA) of over 26 deg, as calculated by the roll rate sensor in the airbag ECU.

Figure 17 shows the RA and Gy values calculated from this sensor in a screwing non-crash type rollover. At the beginning of the event, Gy is generated when the RA increases and reaches approximately 90 deg. In other words, the angle of the surface causes the roll to start. The vehicle then rolls over and Gy is generated when the side of the vehicle impacts the road.

Similarly, Fig. 18 shows a crash type rollover involving a curb trip. Around time-zero when the trigger is generated, RA begins to increase after a sudden spike in Gy. The vehicle starts rolling with the origin of rotation located at the tire, in accordance with the strength of the impact between the tire (the side of the vehicle) and the curb. The initial sudden Gy peak must be accurately identified to determine whether the vehicle will roll over.

Therefore, it is necessary to make separate records for slow non-crash types and sudden crash types of rollover.

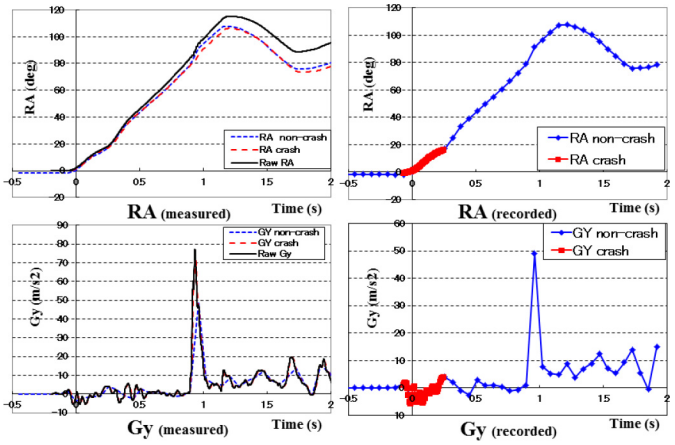


Figure 17. Non-crash type: screwing rollover.

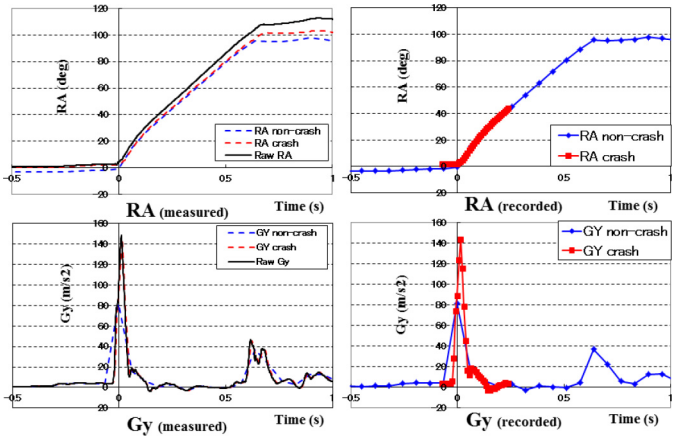


Figure 18. Crash type: rollover after curb trip.

The moving average calculation duration of the recorded data was optimized to record crash and non-crash type rollovers accurately. Results of analysis found that the duration for crash type rollovers should be 8 ms and the duration for non-crash type rollovers should be 64 ms. After optimizing the recording time and amount of recorded data, the number of data samples was set to 40 with a recording time range from -80 ms to 240 ms for crash type rollovers and from -640 ms to 1920 ms for non-crash type rollovers. Judgment whether a rollover is a crash or non-crash type is performed based on the airbag deployment time after the trigger (Table 1).

Table 1. Explanation of buffer data.

Event type	Time to deploy	Data	Number of samples
Crash type	< 240 ms	8 ms moving average	40 10: pre-triggered 30: on/post-triggered
Non-crash type	>= 240 ms or OFF	64 ms moving average	

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It is necessary to separate crash and non-crash type rollovers, which have different record sampling conditions and time periods, and carry out continuous calculation.

1. Moving average of 8 data samples captured over a period of 8 ms from the sampled sensor values at an interval of 1 ms (crash type)
2. Moving average of 8 samples captured over a period of 64 ms using the moving average that was calculated over 8 ms for the crash type rollover (non-crash type)

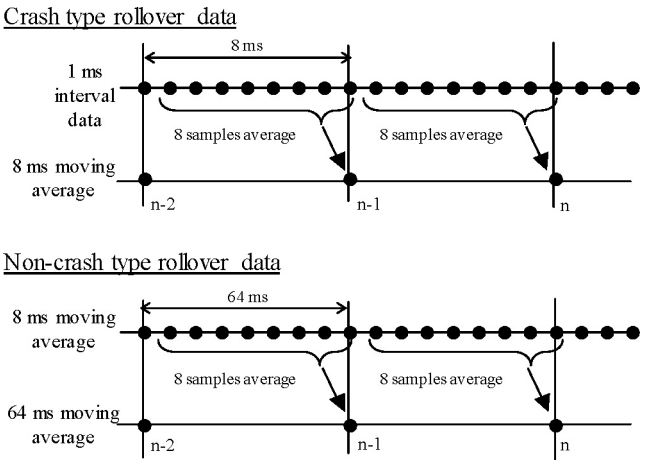


Figure 19. Rollover recorded data.

The event type is determined based on the time from the satisfaction of the rollover trigger judgment conditions to the satisfaction of the airbag deployment judgment conditions.

Either data 1 or 2 is recorded 40 times. The 40 data samples are allocated as follows: 10 before the trigger and 30 after.

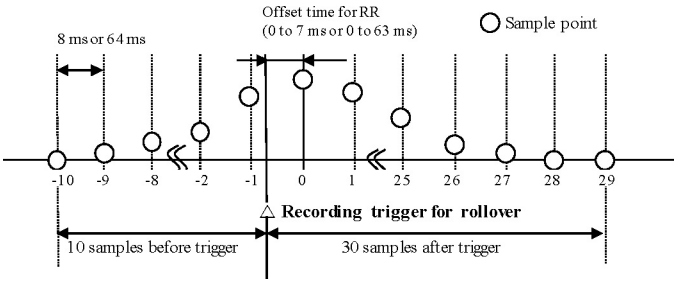


Figure 20. Rollover buffer data (for RA and Gy).

2-2-2-4 Recording of Pop-Up Hood Pedestrian Protection System

The pop-up hood (PUH) system that detects a pedestrian impact and raises the rear end of the hood was introduced in 2012. Fig. 21 shows the configuration of the system. Pedestrian impacts are detected by a chamber sensor installed in front of the bumper. The deformation of the chamber caused by the pedestrian impacting the bumper is detected by two pressure sensors at the left and right. Calculation is performed based on the pressure sensor signals and the vehicle speed. Then the effective mass of pedestrian is calculated. If a pedestrian is detected, an explosive lifter located at the rear end of the hood is fired. The lifter creates a space between the hood and the objects in the engine compartment mainly to protect the head of the pedestrian.

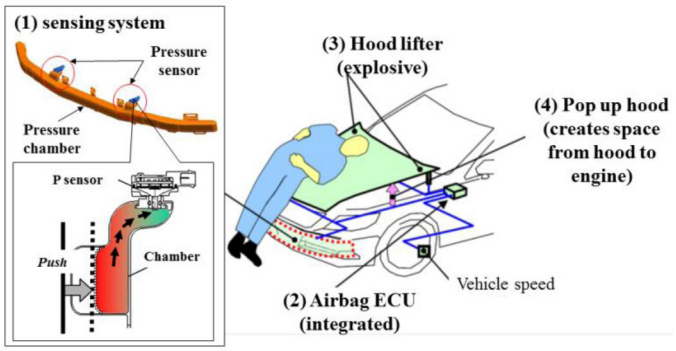


Figure 21. PUH system recorded data.

The ΔP/P value is calculated from the pressure sensor signals after correcting for ambient pressure. This is an effective physical value for detecting the rate of change of pressure (ΔP/P=RP).

To identify the left RP, right RP, and vehicle speed more precisely, the recorded values are calculated using a moving average of the value shown in Fig. 22 sampled 8 times over 4 ms at an interval of 0.5 ms.

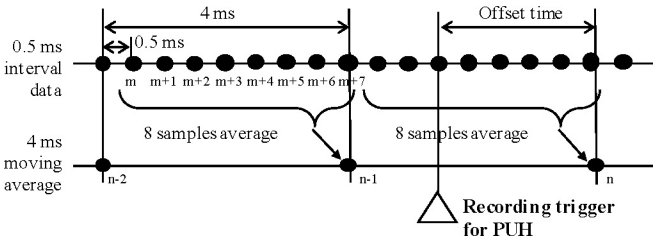


Figure 22. PUH system recorded data.

The left RP, right RP, and vehicle speed values are recorded over a period of 104 ms at an interval of 4 ms.

The trigger is a calibrated value for each vehicle. Each moving average is recorded over the following time range: -24 ms to 80 ms.

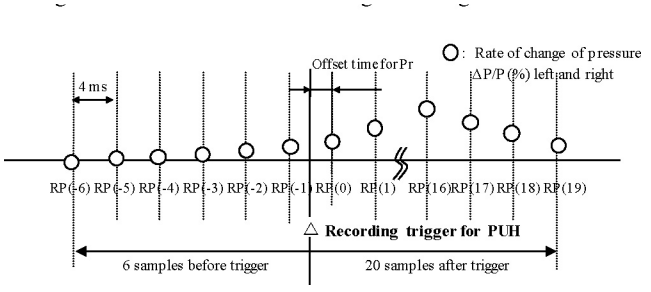


Figure 23. PUH system buffer data (for left RP, right RP, and speed).

APPENDIX B shows an example of PUH system EDR data downloaded from the field.

In addition, if an internal door pressure sensor is used instead of the side crash B-pillar sensor, it should be noted that the recorded value uses a ΔP/P value equivalent to PR and the sampling and recording time follows the description for additional side crash data.

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2-2-3. Recording by Non-Crash Trigger

Since the EDR is capable of recording various events leading up to an accident, non-crash triggers were considered based on field investigations. It was hypothesized that users may experience concern before the accident occurs. Therefore the vehicle control history (VCH) function, which records non-crash trigger conditions, was adopted as a another device other than EDR.

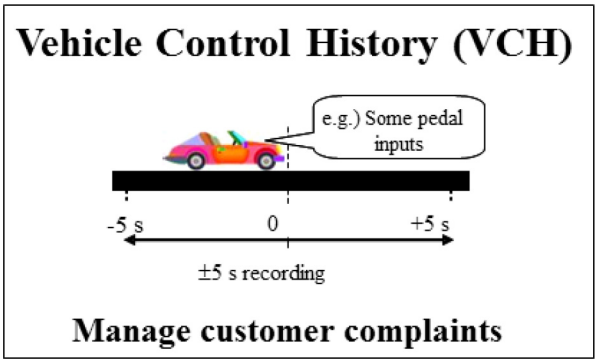


Figure 24. Vehicle control history.

VCH detects triggers generated by certain vehicle behavior or certain driver operations and records the vehicle status. The data is recorded inside the nonvolatile memory of the airbag ECU. VCH data is not erased even if the battery terminals are disconnected.

VCH data is recorded in separate memory areas to EDR. Data is grouped in accordance with the trigger type and recorded in different areas. Up to a total of 140 events can be recorded. Once one area becomes full, the oldest data in that area is over-written by new data.

Trigger time data is captured and recorded by the body ECU (ignition ON count, time elapsed since ignition ON) or the navigation system computer (absolute time, only for factory-installed navigation systems). The TechStream diagnostics tool is used to download the recorded data and recorded data cannot be erased. APPENDIX C shows the VCH trigger conditions.

In addition to certain vehicle behavior and driver operations, operation flags of control systems such as the braking, pre-collision safety (PCS) system, and the like are also included in the trigger conditions.

Sudden braking history is captured using Gx. As a result, driving operations on rough roads or slight crashes can also be recorded, which is not possible with the regulation EDR recording trigger threshold of V150>8 km/h.

APPENDIX D and E show examples of VCH system data downloaded in the field. These are screens from the TechStream tool for a 2014 Lexus IS350. APPENDIX D shows time-based history data and APPENDIX E shows data recorded by the "under TRC operation flag" trigger. It shows the throttle opening due to a sudden accelerator operation, which was suppressed by the traction control.

2-3. Download Tool

The download tool (also called the read-out tool (ROT)) used in the 2010 field investigation was manufactured in-house by Toyota. 230 units of this tool were delivered to distributors around the world from April of that year. This tool was compatible with 1230 different part number airbag ECUs. Subsequently, the U.S. introduced regulations about the download tool, which made it compulsory for data downloading to be achievable using a commercially available tool. Toyota chose the Crash Data Retrieval (CDR) tool manufactured by Bosch, which is regarded as an industry standard and is used by a large number of automakers. By September 2012, this tool could be used to download EDR from all Toyota's models in North America. Currently, CDR is compatible with virtually all of Toyota's EDR-equipped vehicles all around the world.

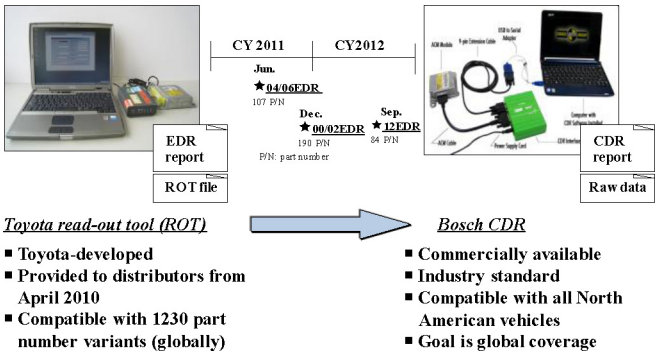


Figure 25. Commercially available tool.

Most automakers, including GM and Ford have also adopted the Bosch CDR data download tool. The BOSCH CDR coverage in vehicles in North America is reported each year at the annual CDR Summit. In January 2015, it was reported that the coverage of Toyota's vehicles was 70%, much higher than the total North American average of 47%.

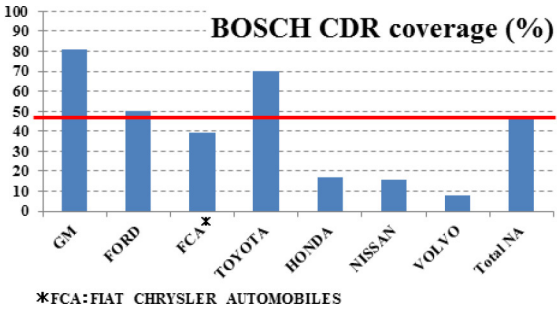


Figure 26. BOSCH CDR coverage in North America.

3. AUTOMATIC COLLECTION OF EDR DATA

In order to build safer vehicles, vehicle development engineers conduct a wide range of crash tests to collect various data. However, actual accidents that occur on the road may not always match the results of these tests, and the amount of information obtained from tests has limits. For these reasons, the collection and analysis of EDR data recorded in the field are extremely important for understanding what happens in actual accidents.

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The following section describes the automatic online EDR data collection system that was constructed in cooperation with Bosch. The objectives of retrieving this EDR data from the field are as follows.

1. To understand the type and severity of events from EDR data
2. To confirm data quality (pre- and post-event)
3. To understand the limitations of EDR data
4. To study events, and to understand crash mechanisms and the effects on vehicle design
5. To identify EDR enhancements for future

Figure 27 shows the system flow. EDR data is downloaded from the vehicle using the Bosch CDR, thereby creating a CDR report. A separate sanitized file is automatically created by deleting personal information from the CDR report. This file is then encrypted. When the CDR PC is connected to the internet, the sanitized and encrypted file is automatically transferred to the Bosch server. The Bosch server then transfers the file to the EDR database of the Collaborative Safety Research Center (CSRC) in North America by a periodical batch process.

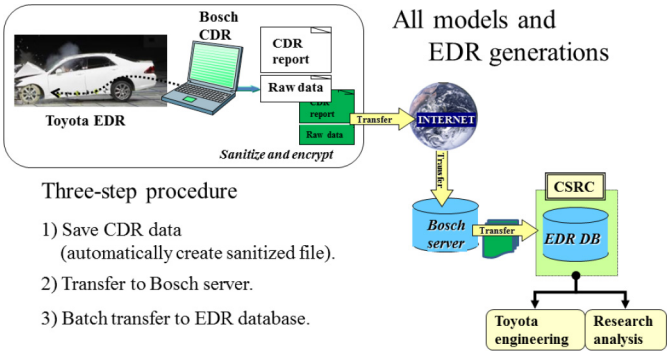


Figure 27. EDR data transfer and storage.

Figure 28 shows how personal information is sanitized. When the CDR report is created, the CDR creates and saves a duplicate file for each CDR readout, removing any personally identifiable information (PII) and the VIN. As the figure shows, the final six digits of the VIN specify an individual. That, and the other individual information entered by hand when the EDR data is downloaded is located in fixed fields in the CDR report header. The crossed out portions of the data are deleted automatically. The readout date and time, as well as the vehicle information and EDR details remain.

CDR File Information — CDR report header

User Entered VIN	4T4BF3EK2AR~~003190~~
User	~~Joe Ol Fe3e~~
Case Number	~~AR003190~~
EDR Data Imaging Date	~~07/02/2012~~
Crash Date	~~05/31/2012~~
Filename	4T4BF3EK2AR~~003190_07022012~~_EDR_ACM.CDRX
Saved on	Monday, July 2 2012 at 11:26:02
Collected with CDR version	Crash Data Retrieval Tool 5.0.2
Reported with CDR version	Crash Data Retrieval Tool 5.0.2
EDR Device Type	Airbag Control Module
Event(s) recovered	Front/Rear (2), Side (2)

Manually inputted at EDR download

Figure 28. Data sanitization.

From March 2013, 14 field engineers from Toyota Motor Sales (TMS) in the U.S. started collecting CDR data. As of January 2015, a total of 300 files had been transferred to the EDR database at CSRC. However, since the transferred files also include vehicle test data unrelated to accidents in the field and duplicated files, a total of 198 valid files have been collected. Although the number of field CDR files received from TMS has been less than originally expected, the effectiveness of the sanitization, file transfer, and conversion and storage functions has been confirmed.

Figure 29 shows the different generations of collected EDR data. The graph separates the data into first (March 2013 to January 2014) and second (February 2014 to January 2015) collection phases. In the first phase, there were many type A (post-crash data only) 02EDRs. More data from the latest 12EDRs and 13EDRs was collected in the second phase.

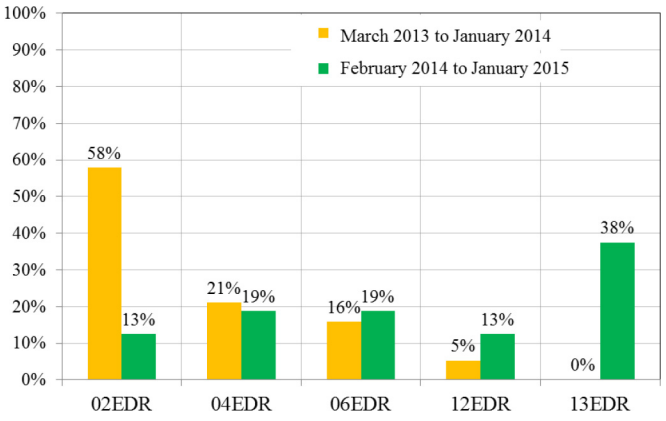


Figure 29. Collected data.

APPENDIX F shows the collected data recorded for a 2014 Prius C. One topic in the collected data is as follows. Figure 30 shows brake depression timing data. Even when drivers brake quickly, braking timings tend to be less than 3 seconds before the crash and the brake pressure reaches a maximum value at the moment of impact. Alerting the driver of the possibility of a crash at an early timing through the PCS system may be an extremely effective way of helping to avoid an accident.

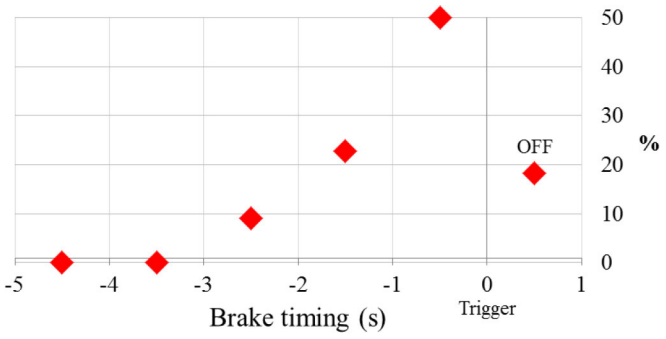


Figure 30. Late brake timing.

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Figure 31 shows the relationship of the braking timing with the crash ΔV and vehicle speed (the last reported vehicle speed before impact). The crash ΔV decreases as the braking timing becomes earlier. In contrast, there is no clear correlation with vehicle speed.

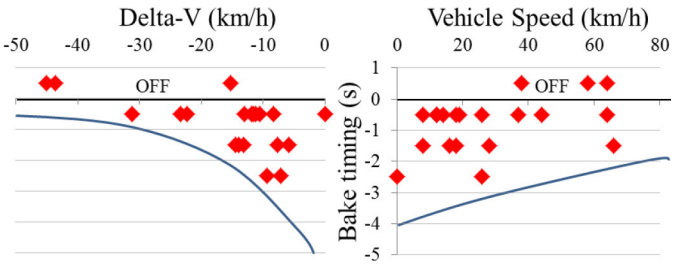


Figure 31. Braking timing vs. ΔV and vehicle speed.

Figure 32 shows the relationship between crash ΔV and vehicle speed. In some cases, crash ΔV increases in accordance with the vehicle speed.

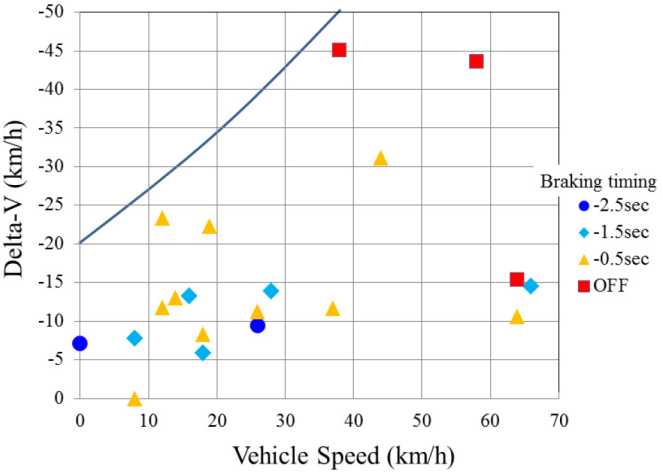


Figure 32. Vehicle speed.

Figure 33 shows the steering operations. In some cases, steering operation data was captured immediately before the crash. The steering angle was less than 90 deg in these cases.

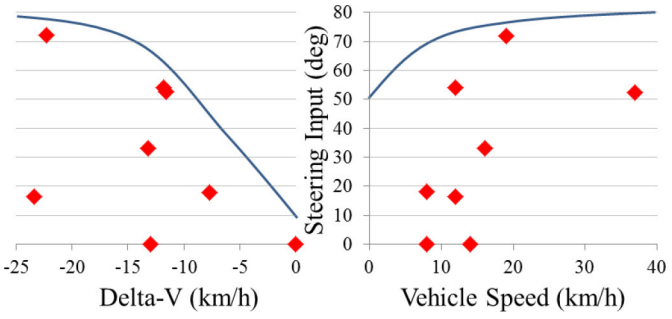


Figure 33. Steering inputs.

Slightly fewer than 200 data groups were collected in around two years. These cases indicate the number of EDR data downloads after a customer request to TMS, not the total of accidents in the field over that period. A large proportion of inquiries about the data of the latest 12EDRs and 13EDRs related to non-deployment of airbags in low-speed crashes. In other words, many customers asked TMS why airbags did not deploy in an accident.

The pre-crash data topic described above differs greatly depending on the accident scenario. The EDR can identify the driver operation in individual cases. After studying the estimated overall statistical population for accidents in the field and the sampling method, it should possible to develop effective safety systems considering human characteristics. One of the objectives of the sanitized EDR data is to estimate these statistical populations through statistical analysis. In addition, by incorporating police reports that contain detailed accident investigation results and data about the severity of driver injuries, it should also be possible to put together an extremely effective database.

Figure 34 summarizes EDR data applications, divided into pre- and post-crash categories. Pre-crash data can be used for active safety systems that function before accidents, while post-crash data can be used for emergency medical services.

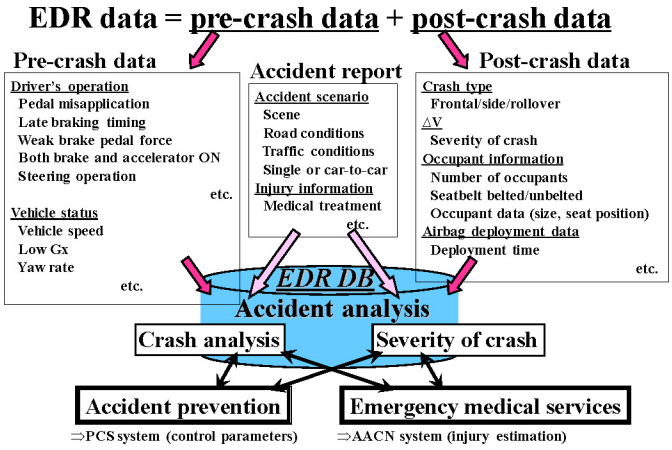


Figure 34. EDR data collection objectives.

Since the property rights of EDR data belong to the owner, it is necessary to respect privacy. With a system that uploads data automatically, it is still important to confirm that EDR information may be collected after deleting personal information from the downloaded EDR data. Figure 35 shows an example of how this is carried out.

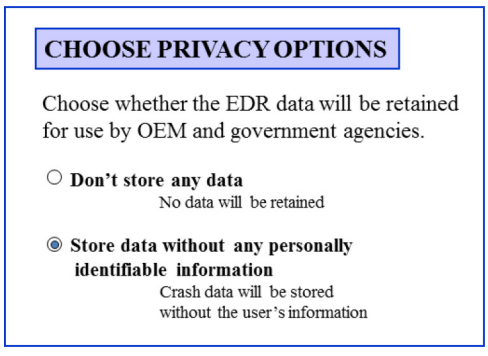
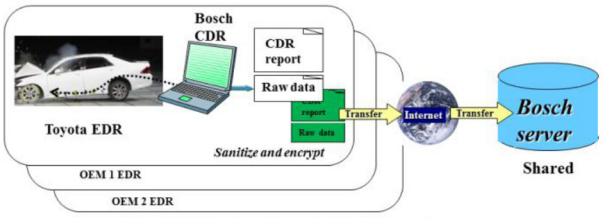


Figure 35. Privacy concerns.

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It would be preferable to expand this trial system to include other automakers. As an industry standard, data could potentially be collected in the Bosch server and shared between people holding a CDR license in the future.



Bosch CDR system to be shared:

1) Install data collection function on CDR (automatically create sanitized and encrypted file).
2) Transfer to Bosch server.
3) Bosch server to be shared with analysts.

Figure 36. EDR data users.

4. TRIAL OF ADVANCED AUTOMATIC COLLISION NOTIFICATION SYSTEM

The new Advanced Automatic Collision Notification (AACN) system that uses EDR data is being studied. AACN functions as follows.

From the post-crash data, ΔV, which expresses the severity of the crash, and seatbelt data is transmitted to a data center. This data is very useful preparatory information for medical services. For example, the center calculates the estimated fatality/serious injury rate from this data and notifies local hospitals and fire departments, which can then dispatch ambulances or medical helicopters based on the rate estimation.

Figure 37 shows an example using the major blood loss that accompanies around one-third of fatal traffic accidents. The curve indicates the relationship between the time after the accident that medical treatment begins and the fatality rate, a relationship that is also known as the golden hour principle. Shortening the time until medical treatment in an accident substantially reduces the fatality rate. The figure shows the actually measured time for medical helicopters to arrive at training accident scenarios in Chiba prefecture, Japan. In this test, medical treatment could be started within 14 minutes of the accident, 24 minutes faster than the average for medical helicopters without the system. This would drastically reduce the accident fatality rate.

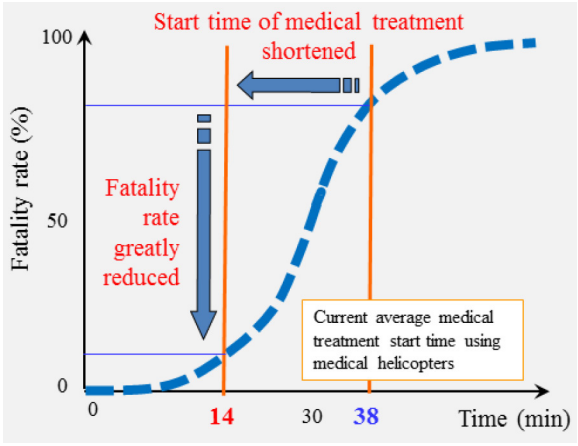


Figure 37. Golden hour principle.

Figure 38 shows how the center calculates the estimated fatality/serious injury rate from the ΔV and seatbelt data [4]. Accident data was extracted from 2.8 million accidents in Japan recorded by the Institute for Traffic Accident Research and Data Analysis (ITARDA). The estimated fatality/serious injury rate at each crash ΔV is calculated for users wearing and not wearing seatbelts. The estimated fatality/serious injury rate is then calculated from these statistically processed risk curves.

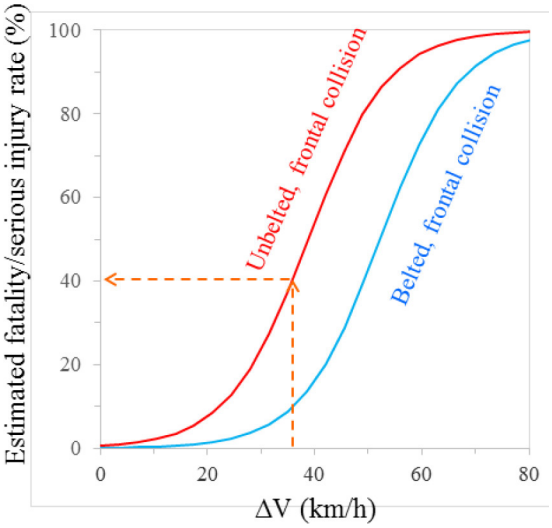


Figure 38. Estimated fatality/serious injury rate.

Figure 39 shows an example of the tablet screen containing information sent from the data center to hospitals and fire departments. It consists of basic vehicle data, the location of the accident, crash direction and ΔV, occupant and seatbelt data, airbag deployment data, as well as the estimated fatality/serious injury rate. The hospitals and fire departments then dispatch medical attention based on this information. The example shown in Fig. 37 above demonstrates the results of training using medical ambulances dispatched using this system.

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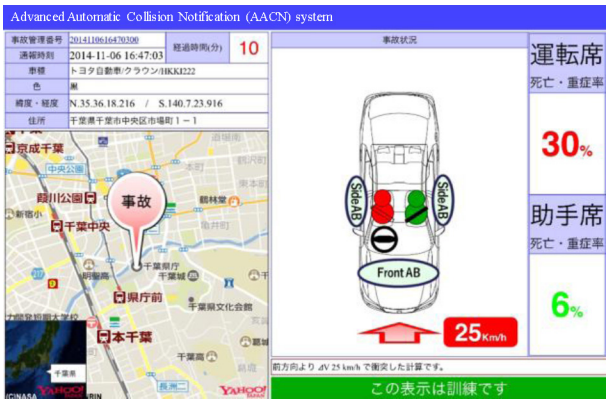


Figure 39. Example of tablet display.

A two-year trial of this system is due to start in November 2015. Nine hospitals are cooperating with the trial (Fig. 40).

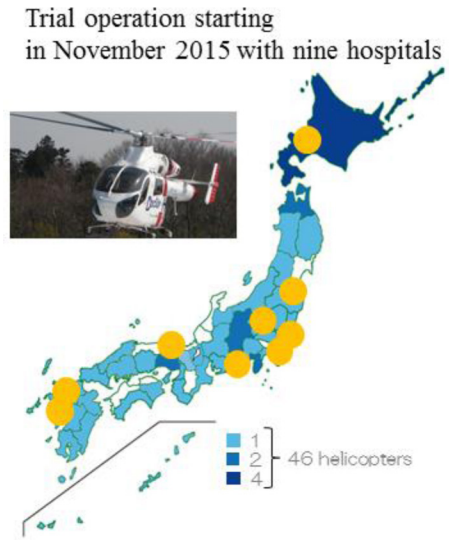


Figure 40. Hospitals cooperating in AACN trial.

5. CONCLUSION

This paper has focused on the unique functions of Toyota's EDRs compared to the requirements described in U.S. EDR regulations. Since an EDR is regarded as an effective analysis tool after an accident has occurred, the goal is to install EDRs on as many vehicles as possible by incorporating the necessary functions into development vehicles at an early stage. Toyota regards the U.S. regulations as minimum requirements and is actively adding additional data capture functions to enable more accurate accident analysis. It is also aiming to optimize data recording methods wherever possible.

PCS and other active safety systems are now being adopted on a wide range of vehicles. The recording of data from these systems is also likely to become more important in the future. Particular key points will be the recording of driver operations and vehicle conditions (including control states) immediately before an accident occurs.

Currently, roughly half of all vehicles in North America are equipped with EDR. In the unfortunate event of an accident, it will be important to save and share data about what happened. Considering privacy concerns, data sharing requires the sanitization of personal information. However, the collection of accident data from the field will play a key role in analyzing accident mechanisms and helping to develop even safer vehicles.

Field tests of the AACN system using EDR data begin in Japan this year. Toyota intends to work closely with the cooperating hospitals and fire departments with the aim of saving lives by speeding up the start of medical treatment.

Research will also continue into expanding the functions of EDRs and the development of new systems using EDR data.

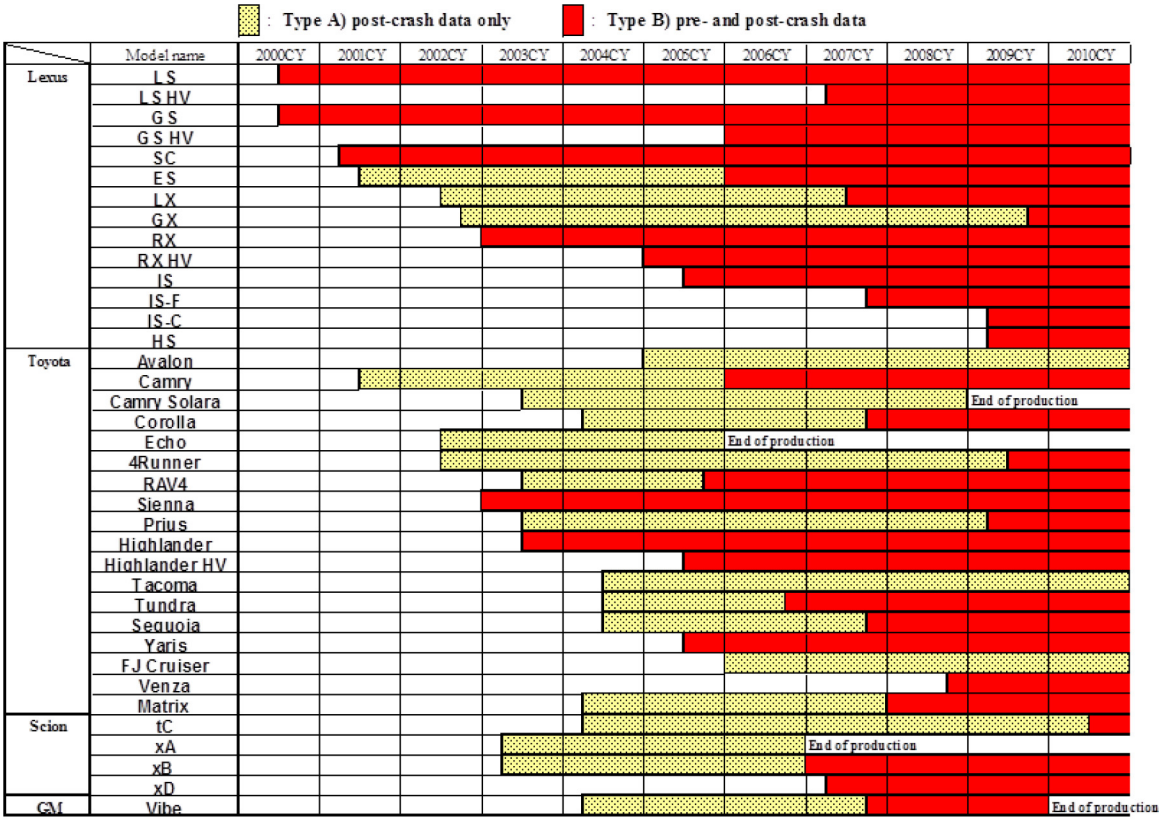
REFERENCES

1. United States Code of Federal Regulations, 49 CFR Part563 "EVENT DATA RECORDERS"
2. Iyoda, M., "Overview of Toyota's Latest EDR," presented at the Highway Vehicle Event Data Recorder Symposium, September 5-6, 2007.
3. Iyoda, M., "Toyota's Latest EDR" (in Japanese), 2008 *Gazette of the General Insurance Association of Japan* 234.
4. Tominaga, S. et al., "Accidental Injury Prediction and Prevention, Medicine - Medical and Engineering Research for Casualties Reduction -," presented at the JSAE Annual Congress (Autumn), Japan.

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APPENDIX

APPENDIX A - EDR Adoption Status in North America



Note 1: The type B EDR for the Tacoma, FJ Cruiser, Avalon, and Scion tC was available by the end of 2010.

Note 2: Data download tool software was completed for all Toyota/Lexus/Scion models by the middle of April 2010.

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APPENDIX B - Downloaded PUH System EDR Data

*** Page Information ***

Recording Status , Time Series Data	Complete
Recording Status, Crash Info.	Complete
TRG Count	1 times
Time from Pre-Crash TRG	0 msec
Linked Pre-Crash Page	Page. 0
Time from TRG to Next Sample	1 msec

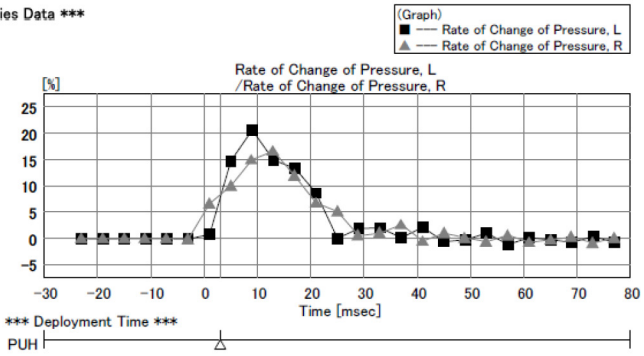
*** Pop-Up Hood Information ***

Establishment Time, Judgment of Pop-Up Hood Operation.	3 msec

*** Time Series Data ***

Time	Calculated Value of Vehicle Speed	Rate of Change of Pressure	
		L	R
[msec]	[km/h]	[%]	[%]
-23	59	0.00	0.00
-19	59	-0.13	0.00
-15	59	0.00	0.00
-11	59	0.00	0.00
-7	59	0.00	0.00
-3	59	-0.13	-0.13
1	59	0.75	6.63
5	59	14.75	10.00
9	59	20.63	15.00
13	59	14.88	16.63
17	59	13.38	11.88
21	59	8.63	6.75
25	59	0.00	5.13
29	59	1.88	0.50
33	59	2.00	1.00
37	59	0.13	2.63
41	59	2.13	-0.50
45	59	-0.63	1.00
49	58	-0.25	0.13
53	58	1.00	-0.63
57	58	-1.13	0.63
61	58	0.13	-0.75
65	58	-0.25	-0.25
69	58	-0.75	0.25
73	59	0.38	-0.88
77	59	-0.75	0.13

*** Time Series Data ***



Rate of change of pressure = ΔP/P (%)

ΔP: pressure change caused by pedestrian

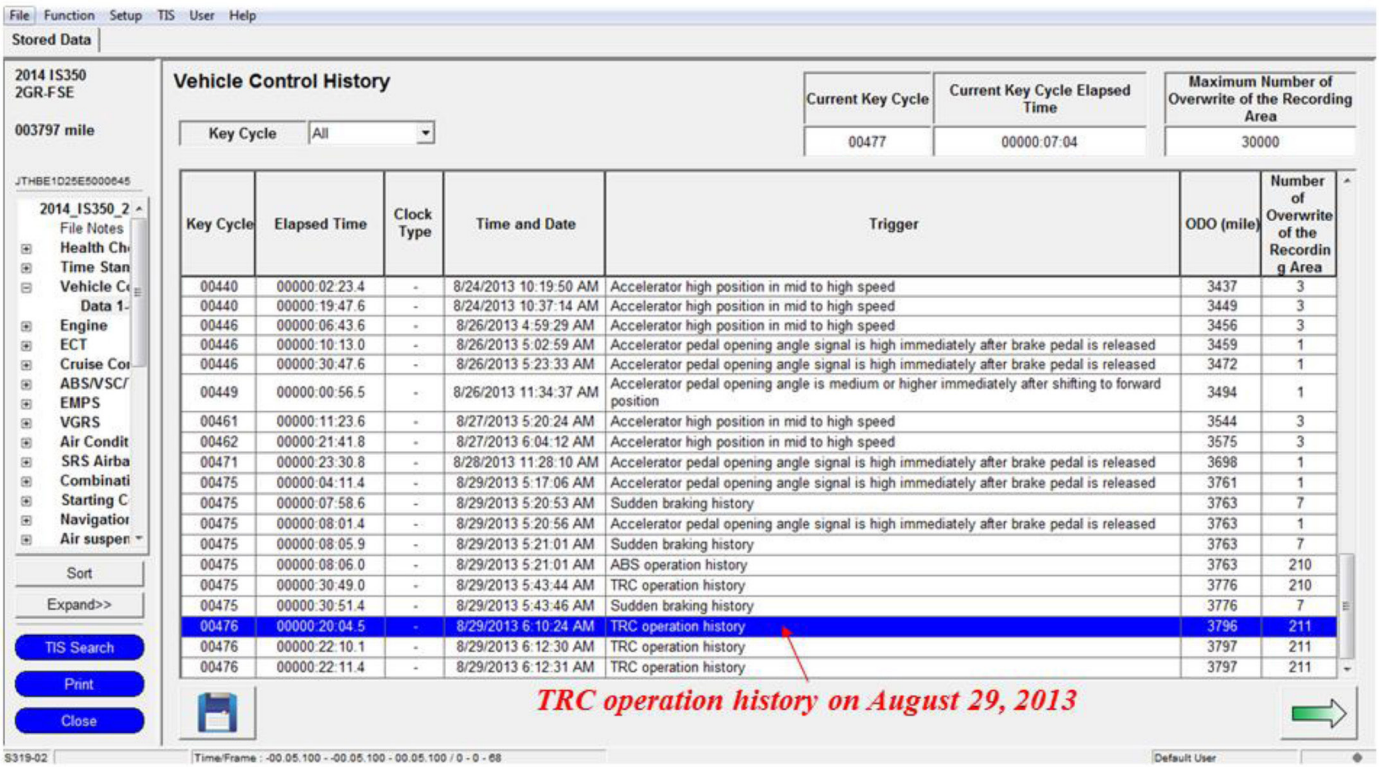
P: ambient pressure

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APPENDIX C - Trigger items of VCH

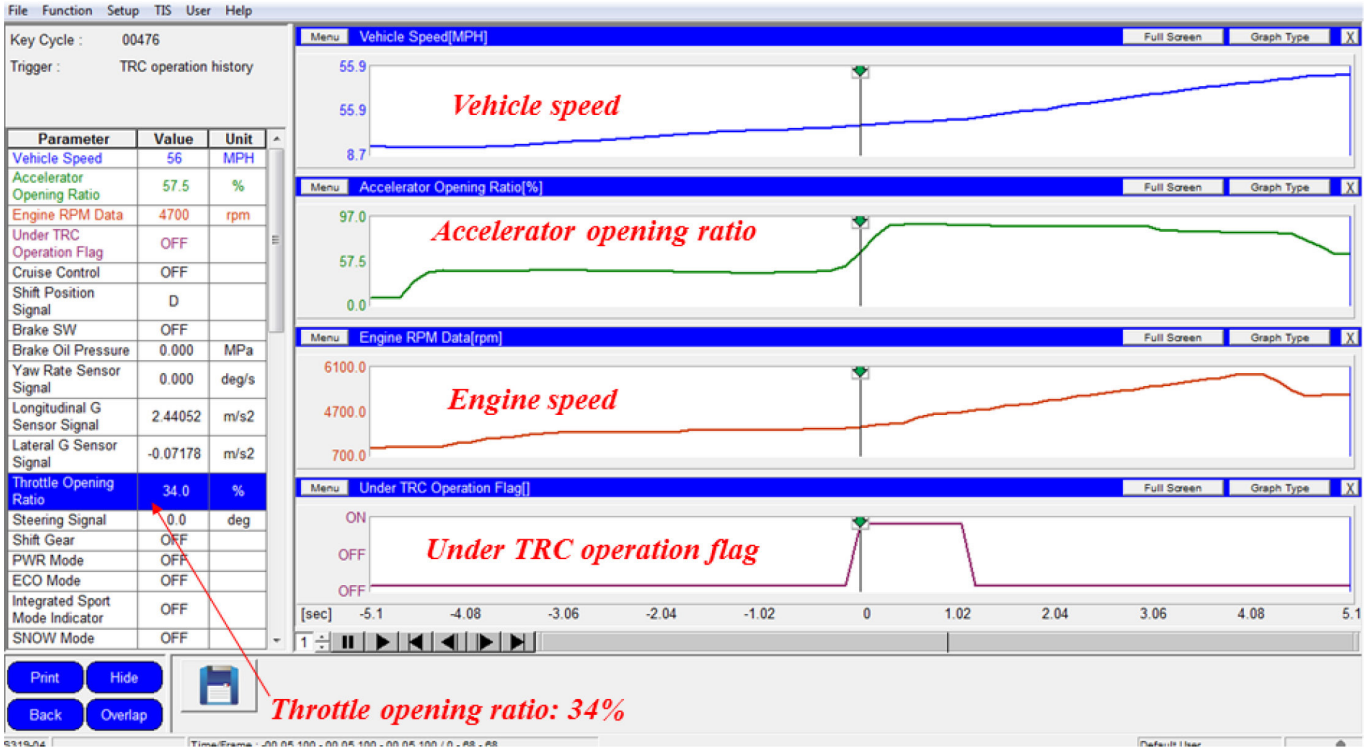
TRIGGER ITEM		RECORD DATA				
		BEFORE DETECTION		AFTER DETECTION		SAMPLING CYCLE (SEC.)
ID NO.	ITEM	RECORDING TIME (SEC.)	NUMBER OF POINTS	RECORDING TIME (SEC.)	NUMBER OF POINTS	
2	Accelerator pedal opening angle signal is high during low speed	7.5	15	7.5	16	0.5
3	Accelerator high position in mid to high speed	7.5	15	7.5	16	0.5
4	Accelerator pedal opening angle signal is high immediately after brake pedal is released	7.5	15	7.5	16	0.5
5-1	Accelerator pedal opening angle is medium or higher immediately after shifting to R	5	10	5	11	0.5
5-2	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position					
5-3	Accelerator pedal opening angle is medium or higher immediately after shifting to driving position					
6-1	R position signal input during medium or higher accelerator signal input	5	10	5	11	0.5
6-2	Forward position signal input during medium or higher accelerator signal input					
6-3	Driving position signal input during medium or higher accelerator signal input					
8	Accelerator signal and brake signal input simultaneously	5	10	5	11	0.5
9	Medium or higher accelerator signal input immediately after switching to D or R	5	10	5	11	0.5
10	Medium or higher accelerator signal input during N	5	10	5	11	0.5
12	VSC operation history	5	34	5	35	0.15
13	TRC operation history	5	34	5	35	0.15
14	ABS operation history	5	34	5	35	0.15
15	Sudden braking history	5	34	5	35	0.15
16	Sudden turning history	5	34	5	35	0.15
17	Accelerator pedal opening angle signal is high during cruise control operation	5	10	5	11	0.5
18-1	PCS operation history (warning buzzer operation)	5	10	5	11	0.5
18-2	PCS operation history (warning brake operation)					
18-3	PCS operation history (pre-collision brake assist operation)					
18-4	PCS operation history (prior brake operation)					
18-5	PCS operation history (pre-collision brake operation)					
18-6	PCS operation history (pre-collision seat belt operation)					
18-7	PCS operation history (when acceleration above certain amount is detected)					
19	Torque sensor signal above certain amount is detected during LKA/LDA operation	8	40	0	1	0.2

APPENDIX D - VCH Data Example (2014 IS)

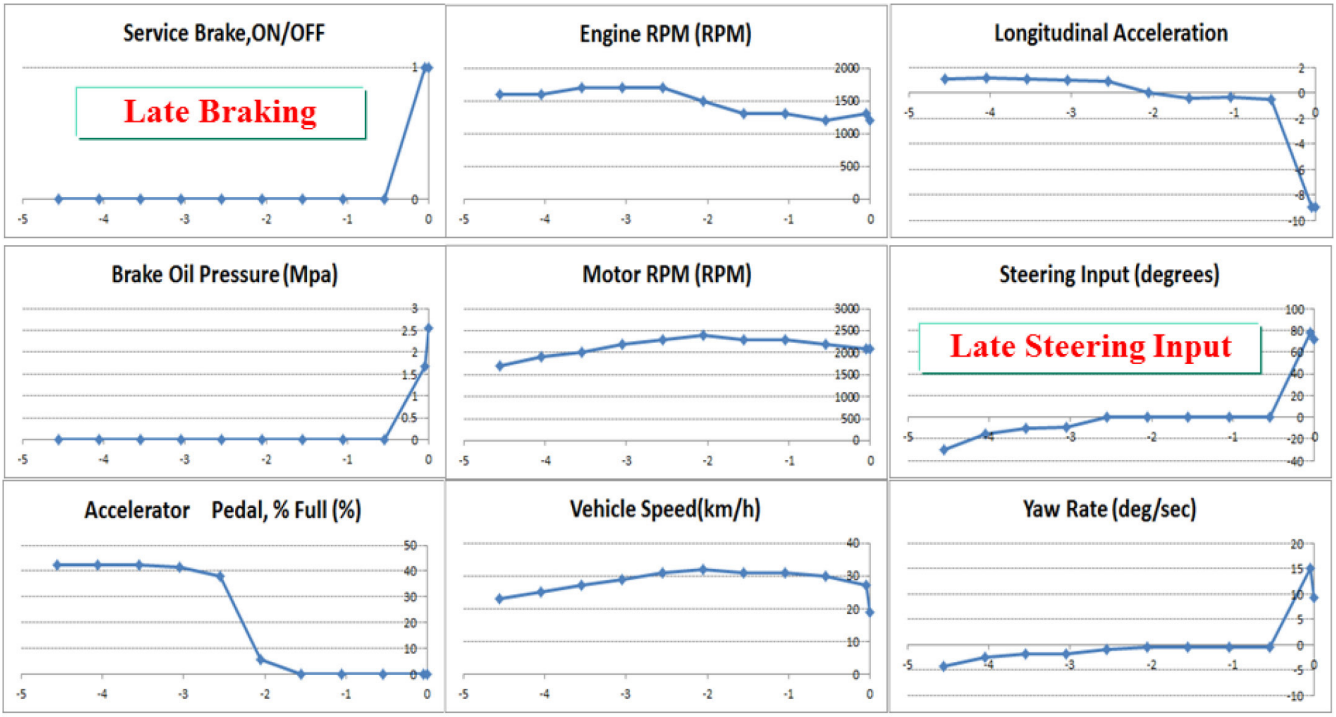


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APPENDIX E - VCH Data Example (2014 IS)



APPENDIX F - Collected EDR Data Example



Prius c : 13EDR front OFF ΔV : -22.3km/h

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